

$$\frac{(0.9)^3}{(0.3)^3} \Rightarrow$$

$$\Rightarrow \left( \frac{0.9}{0.3} \right)^3 \Rightarrow \underline{\underline{27}} \checkmark$$

$$43.75\% \Rightarrow \frac{7}{16}$$

$$6.25\% \times 2$$

$$\downarrow$$

$$\frac{1}{16} \times 2 = \frac{1}{8}$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\begin{array}{c} \text{2}^{(6.4 \times 3)} \\ \text{2}^{19.2} \end{array} \times \text{2}^5 = \text{2}^? \quad \text{3} \checkmark$$

$$\begin{array}{c} \text{2}^{(6.4)} \\ \text{2}^{19.2} \end{array}^3 \div \text{2}^{16 \frac{1}{2}} = \text{2}^? \quad \text{4} \checkmark$$

$$\begin{array}{c} \text{2}^{19.2} \\ \text{2}^{19.2} \end{array} \times \text{2}^5 = \text{2}^? \quad \text{5}$$

$$\begin{array}{c} \text{2}^{19.2+5} \\ \text{2}^{24.2} \end{array} = \text{2}^? \quad \text{6}$$

$$\text{24.2} = ?$$

- (A) 14.20
- (B) 23.60
- (C) 24.20 ✓✓
- (D) 26.40
- (E) None of these

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$2^{5.80} \times 8^{3.20} \times \frac{28}{64^{1.20}} = 2^? \times 112$$

Handwritten work:

$$2^{5.8} \times (2^3)^{3.2} \times \frac{28}{(2^6)^{1.2}} = 2^? \times 112$$

$$2^{5.8} \times 2^{9.6} \times \frac{28}{2^{7.2}} = 2^? \times 112$$

$$2^{5.8 + 9.6 - 7.2} \times 28 = 2^? \times 112$$

$$2^{8.2} \times 28 = 2^? \times 112$$

$$2^{8.2} \times 2^2 \times 7 = 2^? \times 2^3 \times 7 \times 2$$

$$2^{8.2 + 2} \times 7 = 2^{?+5} \times 7 \times 2$$

$$2^{10.2} \times 7 = 2^{?+5} \times 14$$

$$2^{10.2} \times 7 = 2^{?+5} \times 2 \times 7$$

$$2^{10.2} = 2^{?+5+1}$$

$$10.2 = ? + 6$$

$$? = 10.2 - 6 = 4.2$$

(A) 5.20

(B) 6

(C) 6.20 ✓

(D) 6.40

(E) None of these

$$5.8 + 9.6 - 7.2 = x + 2$$

$$5.8 + 2.4 - 2 = x$$

$$6.2 = x$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$512 = 2^9$$

$$1024 = 2^{10}$$

$$\times \frac{1}{1024}$$

$$\text{Something} \times \frac{1024}{1024}$$

$$(2^9)^{7/9}$$

$$512^{7/9} \times 8^{3/2} \div \frac{2}{2048} = 2^? \times 4$$

$$\frac{2^7 \times 2^{4.5} \times 2^{10}}{2^{42}} = 2^{4x}$$

$$2^{(7+4.5+10)} = 2^{(4x)}$$

$$2^{(21.5)} = 2^{(4x)}$$

$$4x = 21.5$$

$$5.375$$

$$(2^3)^{3/2}$$

- (A) 0.225
- (B) 0.250
- (C) 0.375
- (D) 0.125
- (E) None of these

$$5.375$$

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$5^{2.8} \times 5^{-11.8} \times 5^{12} = 5^{2x}$$

$$\begin{array}{c} \uparrow \\ 5^{2.80} \times \underline{25^{-5.90}} \times \frac{1}{\underline{125^{-4}}} = 25^? \\ (5^2)^{-5.9} \quad (5^3)^{-4} \end{array}$$

$$2.8 - 11.8 + 12 = 2x$$

(A) 1  
(B) 1.50 ✓  
(C) 2  
(D) 2.50  
(E) None of these

$$3 = 2x$$

$$\boxed{1.5 = x}$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$3^{2\frac{1}{4}} \times 9^{4\frac{1}{2}} \times 27^{3\frac{1}{4}} = 3^{-?} \times \frac{1}{729^{-3}}$$

$$3^{\frac{1}{2}} \times (3^2)^2 \times (3^3)^{\frac{3}{4}} = 3^{-x} \times \frac{1}{(3^6)^{-3}}$$

$$3^{0.5 + 4 + 2.25} = \frac{3^{-x}}{3^{-18}}$$

$$3^{6.75} = 3^{-x} \times 3^{+18}$$

$$3^{6.75} = 3^{18-x}$$

$$6.75 = 18 - x$$

- (A) 11.25 ✓  
 (B) 12.50  
 (C) 13.25  
 (D) 14.50  
 (E) None of these

$$x = 18 - 6.75$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$7^{5.80} \div 343^{3.20} \div \frac{700}{49^8} = 7^?/100$$

$$7^{5.8} \times \frac{1}{343^{3.2}} \times \frac{49^8}{700} = \frac{7^?}{100}$$

$$7^{5.8} \times \frac{1}{(7^3)^{3.2}} \times \frac{(7^2)^8}{700} = \frac{7^?}{100}$$

$$5.8 - 9.6 + 16 - 1$$

$$7^{11.2} = 7^?$$

(A) 9.60

(B) 10.40

(C) 11.20

(D) 13.60

(E) Cannot be determined

$$? = 11.2$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\begin{aligned} 3^6 &= 729 \\ 3^7 &= 2187 \end{aligned}$$

$$[ ((2187)^{4/5})^{10/12} ]^{2/7} = ?^{1/3}$$

$$3^{\cancel{7} \times \frac{4}{\cancel{5}} \times \frac{10}{\cancel{12}} \times \frac{2}{\cancel{7}}}$$

$$3^{(4/3)} = (x^{1/3})$$

$$3^{\frac{1}{3} \times 4} = (3^4)^{\frac{1}{3}} = x^{\frac{1}{3}}$$

$$3^4 = x \rightarrow 81$$

- (A) 3
- (B) 9
- (C) 27
- (D) 81 ✓
- (E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\begin{aligned} 25^2 &= 625 \\ 25^3 &= 15625 \end{aligned}$$

$$625^? + 5 = 15625^? - 3$$

$$[25^2]^{x+5}$$

$$[25^3]^{x-3}$$

$$25^{(2x+10)} = 25^{(3x-9)}$$

$$2x+10 = 3x-9$$

$$\boxed{19 = x}$$

(A) 1

(B) 4

(C) 8

(D) 13

(E) 19 ✓

What value will come in place of the question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

\*

$$\boxed{5x} + 5$$
$$5[x+1]$$

$$2(? + 5) + 2? = 264$$

$$2^x = 8$$

$$2^{(x+5)} + 2^x = 264$$

$$2^x = 2^3$$

$$\left\{ (2^x \times 2^5) + 2^x \right\} = 264$$

(A) 1

(B) 2

(C) 3 ✓

(D) 4

(E) 5

$$2^x [2^5 + 1] = 264$$

$$2^x [32 + 1] = 264$$

$$2^x \times 33 = 264$$

$$\boxed{x=3}$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\sqrt{x} = x^{\frac{1}{2}}$$

$$\sqrt[3]{x} = x^{\frac{1}{3}}$$

$$\sqrt[4]{x} = x^{\frac{1}{4}}$$

$$\sqrt[5]{x} = x^{\frac{1}{5}}$$

$$x^{?/6} = \sqrt[5]{x \cdot \sqrt{x \cdot \sqrt[3]{x}}}$$

$$\left(1 + \frac{1}{3}\right) = \left(\frac{4}{3}\right)$$

$$\left(x^{\frac{2}{6}}\right) = \left(x^{\frac{1}{3}}\right)$$

$$\left(\frac{?}{6}\right) = \left(\frac{1}{3}\right)$$

$$? = \frac{6}{3}$$

$$? = 2$$

$$x^{\frac{1}{6}} \cdot x^{\frac{1}{3}}$$

$$(A) 2$$

$$(B) 1$$

$$(C) 1/3$$

$$(D) 4$$

$$(E) \text{None of these}$$

$$1 + \frac{2}{3}$$

$$\frac{5}{3}$$

$$x^{\frac{1}{6}} \cdot \left(x^{\frac{1}{3}}\right)^{\frac{1}{2}} = x^{\frac{1}{6} \times \frac{1}{2}} = x^{\frac{1}{12}}$$

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\boxed{216^7} \times 6 \times \frac{36}{\boxed{24^4}} = \frac{6^?}{\boxed{256}}$$

$$6^{21} \times 6^1 \times \frac{6^2}{(6 \times 4)^4}$$

$$\frac{6^{21} \times 6^1 \times 6^2}{6^4 \times 4^4} = \frac{6^?}{4^4}$$

$$21 + 1 + 2 - 4 = x$$

(A) 18

(B) 8

(C) 20 ✓

(D) 24

(E) None of these

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$[(405)^{\frac{1}{2}}]^6$$

$$405^{\frac{1}{2} \times 6}$$

$$(405)^3$$

$$(81 \times 5)^3$$

$$81^3$$

$$(3^4)^3$$

$$3^{12}$$

$$12.50\% \times (\sqrt{405})^6 \div 729^2 \div 125 = (3)^? \div 2\frac{2}{3}$$

$$\frac{1}{8}$$

$$3^{12} \div 3^{12}$$

$$3^{12-12} = 3^{x+1}$$

$$0 = x+1$$

$$x = -1$$

$$(3^6)^2 = 3^{12}$$

$$\frac{3^x \times 3^1}{8}$$

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) None of these

What value will come in place of the question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$0.5 = a$$

$$6 + 2 = x + 3$$

$$\begin{aligned} 0.5 &= \frac{5}{10} \\ (0.5)^2 &= \left(\frac{5}{10}\right)^2 = \frac{25}{100} = 0.25 \\ (0.5)^3 &= \left(\frac{5}{10}\right)^3 = \frac{125}{1000} = 0.125 \\ (0.5)^4 &= \left(\frac{5}{10}\right)^4 = \frac{625}{10000} = 0.0625 \end{aligned}$$

$$(0.5)^6 \times 0.25 = (0.5)^? \times \sqrt{0.015625}$$

$$a^6 \times a^2 = a^? \times [a^6]^{\frac{1}{2}}$$

$$a^6 \times a^2 = a^? \times a^3$$

$$6 + 2 = 3 + ?$$

$$5 = ?$$

(A) 4

(B) 5

(C) 6

(D) 7

(E) None of these

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?



$$\frac{(16 \times 3)^3}{48^3} \div 8^{5/2} \times 256 = 2^? \times 216 \rightarrow 2^7 \times 8$$

$$\frac{16^3}{48^3} \div 8^{5/2} \times 2^8 = 2^? \times 8$$

$$12 - 7.5 + 8 = 2 + 3$$

$$9.5 = x$$

- (A) 6
- (B) 7.50
- (C) 9
- (D) 9.50 ✓
- (E) None of these

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$6^{1.1} \cdot 8^{1.1} = 48^{1.1}$$

$$64.30 \times 368 \times (-0.20) \times (48)^4 \times (2)^{3.3} = 48^?$$

$$4 + (1.1) = ?$$

$$5.1 = ?$$

(A) 4.50

(B) 5.10

(C) 5.60

(D) 6.40

(E) None of these

$$2^{(3 \times 1.1)} = (2^3)^{1.1} = 8^{1.1}$$

What value will come in place of the question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\sqrt[3]{6} \times \sqrt[3]{3} \times \sqrt[4]{10} \times 2 \times \sqrt[5]{125} \Rightarrow 720 \sqrt[60]{6 \times 3 \times 10 \times 5}$$

$$\sqrt{216} \times 3\sqrt{3} \times 4\sqrt{10} \times 2\sqrt{125} = ?$$

$$\begin{aligned} &\sqrt{36 \times 6} \\ &\sqrt{6^2 \times 6} \\ &\sqrt{6^2 \times 6} \\ &6\sqrt{6} \end{aligned}$$

$$\begin{aligned} &720 \sqrt[60]{900} \\ &\sqrt{25 \times 5} \\ &\sqrt{5 \times 5} \end{aligned}$$

- (A) 12500  
(B) 18000  
~~(C) 21600~~  
(D) 27000  
(E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\sqrt[3]{1,17,649} \times \sqrt[3]{46,656} = (?^5 - 940) * 21$$

Handwritten annotations: Blue lines and arrows indicate prime factorization for the cube roots. For 1,17,649, factors 49 and 9 are shown. For 46,656, a factor of 36 is indicated.

$$\sqrt[3]{49} \times \sqrt[3]{36} = ?^5 - 940$$

Handwritten annotations: The cube root of 49 is labeled with a 7, and the cube root of 36 is labeled with a 12. Below the equation, the factors 2 and 3 are written.

- (A) 2
- (B) 3
- (C) 4 ✓
- (D) 5
- (E) None of these

$$84 + 940 = x^5$$

$$1024 = x^5$$

$$4^5 = 1024$$

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\frac{26}{3} \times \frac{24}{7} \times \frac{14}{8} \times \frac{48}{13}$$

$$8\frac{2}{3} \times 3\frac{3}{7} \div \left( \frac{8}{14} \div 3\frac{9}{13} \right) = ?$$

$$192 = ?$$

- (A) 96
- (B) 192 ✓
- (C) 84.50
- (D) 90
- (E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$4\frac{3}{4} \div \sqrt{9025} \times 7\frac{3}{5} \% \text{ of } 26\frac{6}{19} = ? \% \text{ of } 25$$

$$\frac{19}{4} \times \frac{1}{95} \times \frac{38}{500} \times \frac{500}{19} = \frac{2}{1}$$

- (A) 38
- (B) 40
- (C) 45
- (D) 95
- (E) None of these

What value will come in place of the question mark (?) in the following question?  
 निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$25\% \text{ of } \left(68 + \frac{x}{5}\right) = \left(17 + \frac{1}{5}\right)$$

$$68 \frac{x}{5} + 17.2$$

$$68 \frac{?}{5} \% \text{ of } 25 = 4 \frac{8}{9} + 3 \frac{2}{5} - 2 \frac{1}{5} + 11 \frac{1}{9}$$

$$= 16 + 1 + \frac{1}{5}$$

$$25\% \text{ of } \left(68 + \frac{x}{5}\right) = 17 + \frac{1}{5}$$

$$\frac{1}{4} \times \frac{x}{5} = \frac{1}{5}$$

$$x = 4$$

(A) 1

(B) 2

(C) 3

(D) 4

(E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$\underline{6.25\% \text{ of } 1} \div \underline{0.05} = ? \div 4$$

$$\frac{1}{16} \times \frac{1 \times 20}{1}$$

$$\frac{20}{16}$$

$$\frac{5}{4} = \frac{?}{4}$$

$$\textcircled{? = 5}$$

(A) 2.5

(B) 10

(C) 20

(D) 5 ✓

(E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$12.5\% \text{ of } 72 - 7.14\% \text{ of } 56 = ? \times \left(1 - 33.33\%\right)$$

$$9 - 4 = x \left[ \frac{2}{3} \right]$$

$$5 = \frac{2x}{3}$$

$$\frac{15}{2} = x$$

(A) 15

(B) 2.5

(C) 5

(D) 10

(E) None of these

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$8.33\% \times 7$$

$$58.33\% \text{ of } ? - 31.25\% \text{ of } 80 = 31$$

$$\frac{7x}{12} - 25 = 31$$

$$\frac{7x}{12} = 56$$

$$x = 96$$

$$6.25\% \times 5 = \frac{5}{16}$$

$$\frac{5}{16} \times 80 = 25$$

- (A) 48
- (B) 60
- (C) 120
- (D) 96 ✓
- (E) None of these

What value will come in place of the question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$(41.67\% + 87.50\%) \times 120 = 140 + 300\% \text{ of ?}$

- (A) 48
- (B) 60
- (C) 120
- (D) 96
- (E) None of these

$? = 5$

What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$6.6\bar{6} \times 2 \\ \Rightarrow 2/15$$

$$24 + 13.33\% \times (\sqrt{625} + ?) = \sqrt{1024}$$

$$\left(\frac{2}{15}\right) [25 + x] = (8)^4$$

$$x + 25 = 60$$

$$x = 35$$

- (A) 20
- (B) 75
- (C) 10
- (D) 35 ✓
- (E) None of these

What value will come in place of the question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$$(17 - \underbrace{6.67\%}_{\left(\frac{1}{15} \times 30\right)} \times \underbrace{30}_{(42 - 30)}) \div (42 - \underline{\underline{83.33}}) \times 36 = \frac{100}{20} \% \text{ of } 5$$

*Handwritten notes:*  
Top right:  $\frac{5}{6} \times 36 = 6$  and  $100 - 16.66 = 1 - \frac{1}{6}$

$$15 \div 12$$

$$\frac{15}{12}$$

$$\frac{5}{4} = \frac{x}{20} \Rightarrow x = 25$$

$$\boxed{25 = x}$$

- (A) 20
- (B) 37.50
- (C) 26.67
- (D) 25 ✓✓
- (E) None of these